MILITARY SPECIFICATION

METER, SALINITY INDICATING, ELECTRIC, FLUSH MOUNTING RECTANGULAR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of salinity indicating meter suitable for use with salinity indicating equipment specified in MIL-S-15103.

2. APPLICABLE DOCUMENTS

2.1 Issue of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116 - Preservation, Methods of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, Code ATT, 2800 South 20th Street, Philadelphia, Pa. 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-S-901 - Shockproof Equipment, Class HI, (High Impact), Shipboard Application Tests for.
MIL-E-917 - Electric Power Requirement; Equipment for (Naval Shipboard Use).
MIL-S-933 - Screws, Machine, Cap and Set; and Nuts.
MIL-I-983 - Interior Communitcation Equipment, Naval Shipboard; Basic Design Requirements for.
MIL-E-2036 - Enclosures for Electric and Electronic Equipment (Naval Shipboard Use).
MIL-S-15103- Salinity Indicating Equipment.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-108 - Definitions of Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402).

3. REQUIREMENTS

3.1 Materials. The material for each part shall be as specified herein. In general, material requirements shall conform to the applicable requirements of MIL-E-917 unless otherwise specified herein. When a definite material is not specified, a material shall be used which is adequate for the purpose intended and for meeting the requirements specified herein. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

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3.2 Design and construction. The salinity indicating meters shall be of the power factor type electrically designed for use with salinity indicating equipment specified in MIL-S-15103. The meter shall be rectangular in shape, shall have a meter scale not less than 3.0 inches long, measured on the arc, and shall be designed for continuous indication. The meter scale shall be graduated in equivalent parts per million (EPM) of chloride from 0 to 1.5 with zones of the scale colored as specified herein.

3.2.1 Case. The meter case shall be made of molded thermosetting plastic material. The case shall be of drip-proof construction in accordance with MIL-E-2036 and MIL-STD-108. The dimensions of the meter case shall be as specified in figure 1.

3.2.1.1 Window. The meter case shall be provided with a window of glass, or other transparent material, free from discoloration, scratches, striae, electrostatic effect or other defects which may interfere in any way with normal reading of the meter in daylight. The window shall be secured by clamps, or by other suitable means. Cement alone shall not be considered sufficient for holding the window in place.

3.2.1.2 Mounting flange. The case shall have a mounting flange containing four size 6-32 threaded studs for mounting the meter to the salinity indicating panel. The studs shall be fabricated of brass and shall have an exposed length of 3/4 ± 1/32 inch. Each stud shall be supplied with one nut and one lockwasher, fabricated of brass or stainless steel, for fastening the meter to the salinity indicating panel. Spacing of the studs shall be as shown in figure 2.

3.2.2 Dial plate. The meter dial plate shall be fabricated of brass or aluminum, shall be firmly secured to its mounting and shall be readily removable. The dial plate shall be of sufficient thickness and so secured to insure freedom from warping. The dial plate shall be finished with a white matte background with black figures.

3.2.2.1 Scale. The meter scale shall be graduated in equivalent parts per million (EPM) of chloride from 0 to 1.5 EPM with zones of the scale colored in accordance with the following:

Scale reading	Color
0 to 0.065 EPM	WHITE
0.065 to 0.13 EPM	GREEN
0.13 to 1.5 EPM	RED

The lower part of the scale shall contain EPM designations as follows: 0, 0.02, 0.04, 0.06, 0.08, 0.1, 0.15, 0.2, 0.3, 0.4, 0.5, 1.0 and 1.5 EPM, with distinctive markings in the upper part of the scale at 0.065, 0.13, 0.261 and 1.305.

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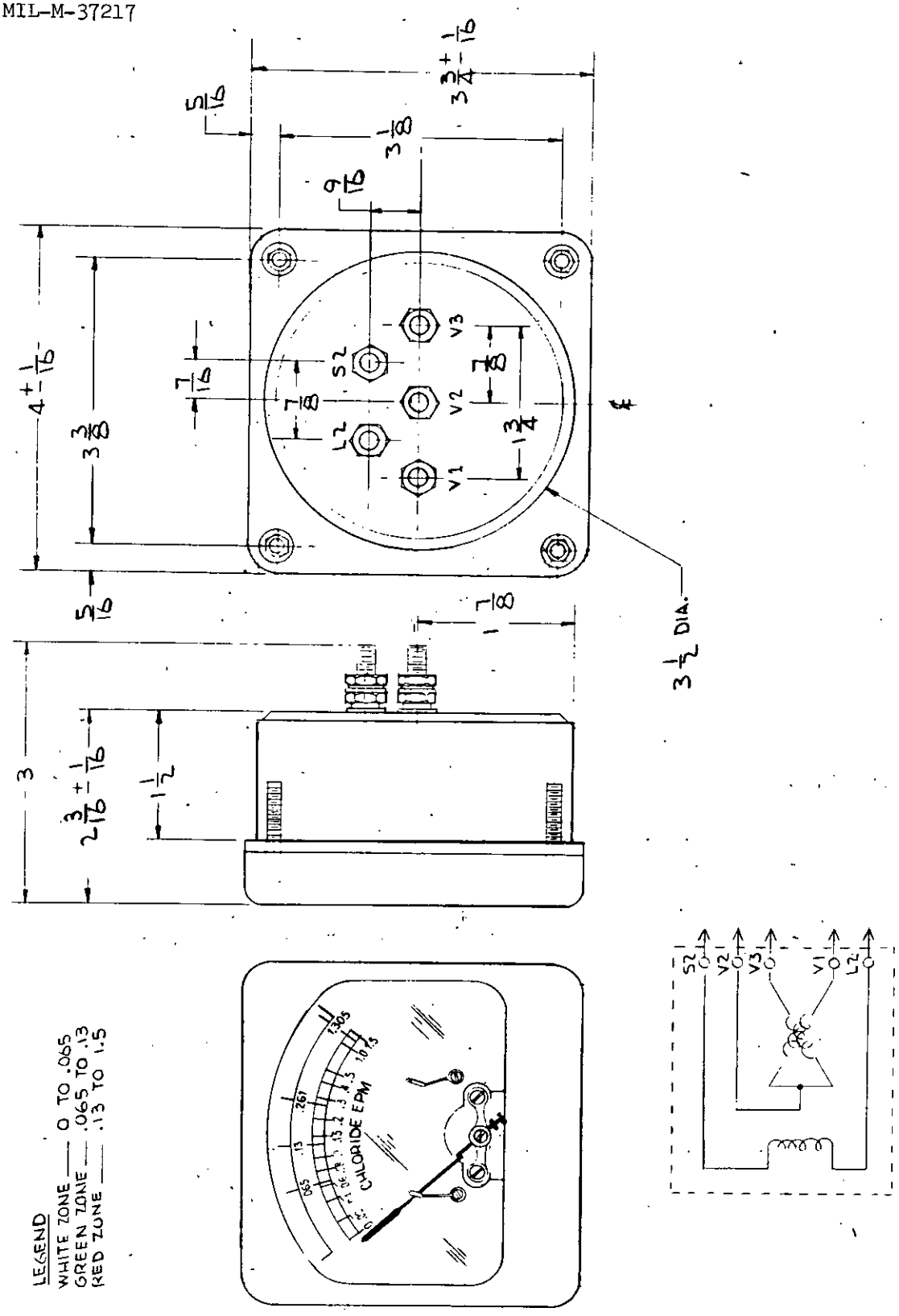


FIGURE 1. Meter salinity indicating.

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3.2.3 Pointer. The meter shall have a pointer formed to permit accurate readings from a distance of two feet. The pointer shall be of rigid construction and suitable for the intended application, and shall not become damaged or distorted when subjected to any of the tests specified herein. The meter shall be equipped with stops, one at each end of scale, to limit the travel of the pointer beyond the scale ends.

3.2.3.1 Pointer clearance (parallax). Where the meter is not provided with means for avoiding errors in reading due to parallax, the clearance between the pointer index and the graduated scale shall not exceed 0.10 inch, nor be less than 0.03 inch.

3.2.4 Scale visibility. The scale and window shall be so designed that when the dial is vertical and the entire scale of the meter can be easily read at any angle up to 45 degrees below the horizontal centerline, with general illumination coming from above and in front and at an angle 45 degrees from the horizontal centerline. Under these conditions there shall be no objectionable shadow on the graduated scale.

3.2.5 Terminal Studs. Five 10-32 NF threaded terminal studs shall be firmly anchored to the rear of the meter in such a manner as to prevent turning or movement. Each terminal stud shall be furnished with two nuts, a flat washer and a lock washer of stainless steel or brass. The terminals shall be permanently and legibly marked as specified in figure 1. The terminals shall be fabricated from brass rod stock and have an exposed threaded length of 3/4 ± 1/32 inch.

3.2.6 Adapter plate. The meter shall be supplied with one of each plate shown in figure 2. The adapter plates shall be fabricated from 0.060 to 0.090 inch thick aluminum alloy. The adapter plates shall have an anodic or iridite coating for protection against corrosion. One side and the edges or both sides and the edges of the plates shall have a black, semi-gloss or gray baked enamel finish. The adapter plate shall be used for mounting the meter to the salinity panel.

3.2.7 Threaded parts. All threaded parts shall be in accordance with Handbook H28.

3.2.8 Hardware. All screws, nuts, washers, and lockwashers shall be in accordance with MIL-S-933.

3.2.9 Locking of screw-thread assemblies. All screw-thread assemblies shall be made vibration-proof. When practicable, split-type lockwashers or equivalent means shall be provided under all nuts.

3.3 Calibration. The meter shall be calibrated to within ± 2 percent when tested as specified in 4.4.1.

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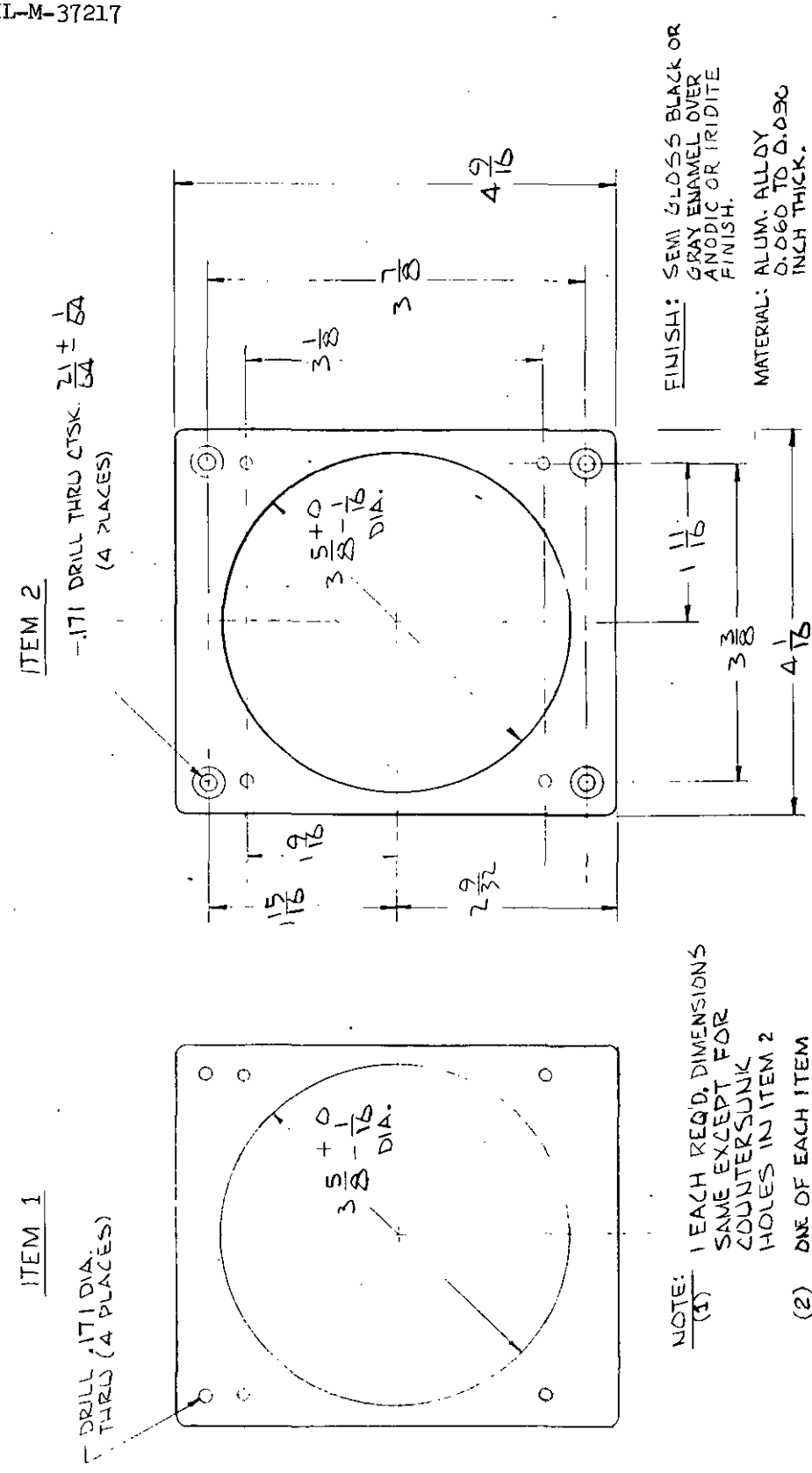


FIGURE 2 - Adapter Plate

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3.4 Shock. The meter shall meet the calibration requirements of 3.3 and shall show no damage or misalignment of parts when tested as specified in 4.4.2.

3.5 Vibration. The meter shall meet the calibration requirements of 3.3 and shall show no damage or misalignment of parts when tested as specified in 4.4.3.

3.6 Dielectric strength. The meter shall withstand the application of 900 volts, 60Hz for a period of one minute without breakdown when tested as specified in 4.4.4.

3.7 Magnetic influence. The meter shall meet the calibration requirements of 3.3 when tested as specified in 4.4.5.

3.8 Accuracy. The meter shall be accurate to within ± 5 percent when tested as specified in 4.4.6.

3.9 Identification marking. The meter shall be permanently and legibly marked with the name or registered trademark of the manufacturer.

3.10 Service data. The contractor shall enclose, with each unit, two copies each of one or more sheets which contain complete instructions for installation. The instruction sheet shall include wiring diagram.

3.11 Workmanship. Workmanship shall be first class throughout. Item shall be free from defects which detract from its appearance or may impair its serviceability. All burrs and sharp edges shall be removed from the item.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

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4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the materials may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or the supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.1.4 Certification in lieu of testing. Certification in lieu of testing (4.4.2 and 4.4.3) is acceptable provided that the contractor certifies in writing that the raw materials, processing and fabrication are identical to those used in meters previously tested and found to be in compliance with the requirements specified herein. The contractor shall furnish the certification to the Defense Personnel Support Center, Technical Services Branch, Directorate of Medical Materiel via the cognizant Government representative.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table I. The unit of product for sampling purposes shall be one meter.

TABLE I. Sampling for examination.

Examination	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	4.0

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and table II. The unit of product for test purposes shall be one meter. The acceptance number shall be zero for all sample sizes.

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TABLE II. Sampling for tests.

Unit of product	Characteristic	Requirement	test procedure	Inspection level
Meter	Calibration	3.3	4.4.1	100%
Meter	Shock	3.4	4.4.2	S-2
Meter	Vibration	3.5	4.4.3	S-2
Meter	Dielectric strength	3.6	4.4.4	S-2
Meter	Magnetic influence	3.7	4.4.5	S-2
Meter	Accuracy	3.8	4.4.6	S-2

4.3 Examination.

4.3.1 Meter. The meter shall be examined for defects including, but not limited to, those listed in table III.

TABLE III. Classification of defects.

Category	Defect
Major:	
101	Component parts missing.
102	Component parts loose or ill fitting.
103	Interior of meter not free of foreign material which might interfere with the normal operation of the meter.
104	Window not free of discoloration, scratches, cracks, breaks or foreign matter.
105	Window not sufficiently transparent.
106	Scale markings not legible.
107	Case not free of cracks or breaks.
Minor:	
201	Identification marking not complete, correct, legible and permanent.
202	Case not free of scratches, rough edges, dirt, grease or foreign matter.

4.3.2 Dimensions. The meter shall be examined for defects in dimensions. Any dimension not within the tolerance specified in figure 1 shall be classified as a defect.

4.3.3 Packaging inspection. The inspection of the preservation-packaging and interior package marking shall be in accordance with the group A and B quality conformance inspection requirements of MIL-P-116. The inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

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4.4 Tests. Tests shall be conducted to determine compliance with requirements specified herein.

4.4.1 Calibration. The meter shall be placed in an electric circuit as specified in figure 3. A resistance decade box, or similar equipment, shall be used to obtain the resistance (R_X) specified in table IV. The meter shall indicate the equivalent parts per million (EPM) associated with the resistance value (R_X) specified in table IV.

TABLE IV. Resistance at given EPM of chloride.

EPM	R_X(ohms)	EPM	R_X(ohms)
0.00	-	0.20	1,894
0.02	18,940	0.261	1,451
0.04	9,470	0.30	1,263
0.06	6,313	0.40	947
0.065	5,828	0.50	758
0.08	4,735	1.00	378.8
0.10	3,788	1.305	290
0.13	2,914	1.50	252
0.15	2,525		

4.4.2 Shock. The meter shall be mounted in a salinity indicating panel which meets the requirements of MIL-S-15103. The shock test shall be in accordance with MIL-S-901 for type C, lightweight equipment.

4.4.3 Vibration. The meter shall be mounted in a salinity indicating panel which meets the requirements of MIL-S-15103. The vibration test shall be in accordance with MIL-STD-167 for type I, environmental vibration, except that the variable frequency test shall be omitted.

4.4.4 Dielectric strength. The dielectric strength test shall be conducted in accordance with MIL-I-983 for 120 volt, 60Hz equipment.

4.4.5 Magnetic influence. The meter shall be mounted on a steel panel of not less than 0.0747 inch thick, and shall be tested in accordance with the method specified in 4.4.1. The meter shall be tested at five points on the EPM scale.

4.4.6 Accuracy. The meter shall be tested in accordance with the method specified in 4.4.1. The meter shall be tested in the following five positions; vertically, tipping the meter 45° forward, 45° backward, 45° to the left and 45° to the right of vertical. A different point on the EPM scale shall be selected for each position of the meter.

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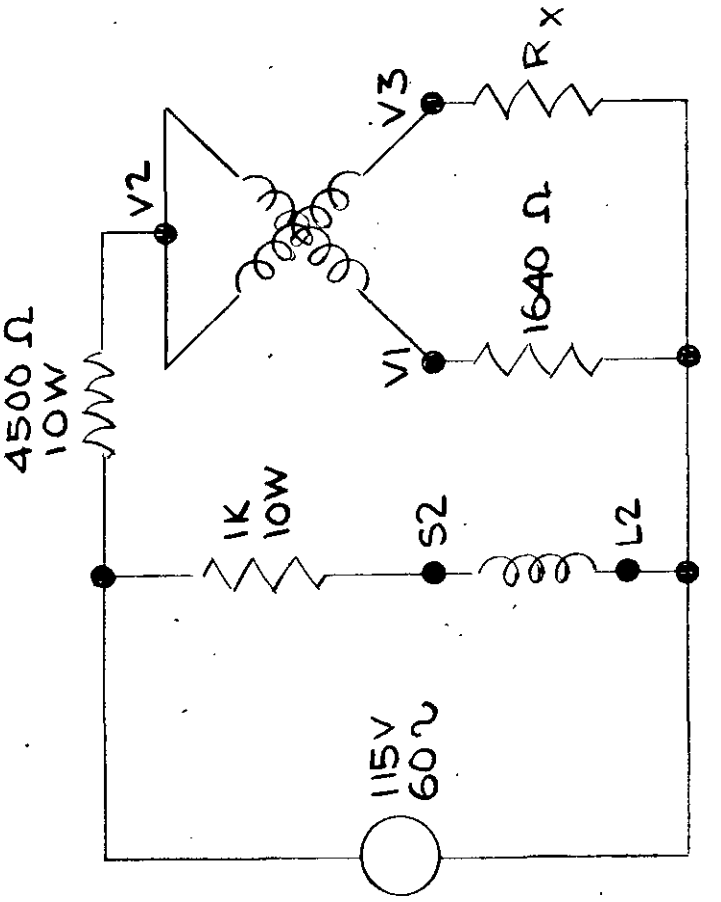


FIGURE 3. Test circuitry

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5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. Each meter shall be thoroughly cleaned and dried in accordance with the applicable method as outlined in MIL-P-116.

5.1.1.2 Preservative. Preservative shall not be applied.

5.1.1.3 Unit packaging. Each meter shall be wrapped in neutral or other suitable anticorrosive materials. Each meter shall be further packaged in accordance with MIl-P-116, method 1C-2.

5.1.2 Level C. Each meter shall be clean, dry and individually packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Six (6) meters shall be packed in a box designed for a type 2 load, and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2 or PPP-B-636, class weather resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.2 Level B. Six (6) meters shall be packed in a box designed for a type 2 load, and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The packaged meters shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations.

5.3 Marking. Each unit package and exterior container shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Intended use. The salinity indicating meters covered by this specification is intended for use in salinity indicating equipment for naval shipboard use.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(c) The National Stock Number (NSN).

6.3 This specification is intended to cover the following item listed in the Federal Supply Catalog.

National stock number	Item identification
6630-00-527-7046	METER, SALINITY INDICATING, ELECTRIC, FLUSH MOUNTING, RECTANGULAR.

CUSTODIANS:
Navy - MS

Preparing activity:
DSA-DM

Project No. 6630-0243

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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DOCUMENT IDENTIFIER AND TITLE
MIL-M-37217 Meter, Salinity Indicating, Electric, Flush Mounting Rectangular

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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